EE1101 Signals and Systems APR-JUN 2021
Tutorial 4

1. Determine whether the LTI systems with following impulse responses is causal and/or stable. Justify your answers.
   (a) $h[n] = (-\frac{1}{2})^n u[n] + (1.01)^n u[1-n]$
   (b) $h[n] = n(\frac{1}{3})^n u[n-1]$
   (c) $h(t) = e^{2t}u(-1-t)$
   (d) $h(t) = e^{-6|t|}$
   (e) $h(t) = (2e^{-t} - e^{(t-100)/100})u(t)$

2. For each of the following impulse responses,determine whether the corresponding system is (i)memoryless, (ii) causal, and (iii)stable.
   (a) $h(t) = \cos(\pi t)$
   (b) $h(t) = e^{-2t}u(t-1)$
   (c) $h(t) = u(t+1)$
   (d) $h(t) = 3\delta(t)$
   (e) $h(t) = \cos(\pi t)u(t)$
   (f) $h[n] = (-1)^n u[-n]$
   (g) $h[n] = (1/2)^{|n|}$
   (h) $h[n] = \cos(\pi n/8)(u[n] - u[n-10])$
   (i) $h[n] = 2u[n] - 2u[n-5]$
   (j) $h[n] = \sin(\pi n/2)$
   (k) $h[n] = \sum_{p=-1}^{\infty} \delta[n-2p]$

3. Evaluate the step response for the LTI systems represented by the following impulse responses:
   a) $h[n] = (-1/2)^n u[n]$
   b) $h[n] = nu[n]$
   c) $h(t) = e^{-|t|}$
   d) $h(t) = (1/4)(u(t) - u(t-4))$
   e) $h(t) = u(t)$

4. Determine whether each of the following statements concerning LTI systems is true or false. Justify your answers.
   (a) If $h(t)$ is the impulse response of an LTI system, and h(t) is periodic and nonzero, the system is unstable.
   (b) The inverse of a causal LTI system is always causal.
   (c) If $|h[n]| < K$ for each n, where K is a given number, then the LTI system with $h[n]$ as its impulse response is stable.
   (d) If a discrete-time LTI system has an impulse response $h[n]$ that is bounded and of finite duration, the system is stable.
   (e) If an LTI system is causal, it is stable.
   (f) The cascade of a noncausal LTI system with a causal one is necessarily noncausal.
   (g) A continuous-time LTI system is stable if and only if its step response $s(t)$ is absoultely integrable, that is,
   $$\int_{-\infty}^{+\infty} |s(t)|dt < \infty$$
   (h) A discrete-time LTI system is causal if and only if its step response $s[n]$ is zero for $n < 0$.

5. Consider two systems A and B. It is given that system A is LTI and system B is an inverse of system A.
   (a) Prove that system B is linear.
   (b) Prove that system B is time-invariant.

6. The autocorrelation function of a signal $x(t)$ is:
   $$\phi_{xx}(t) = \int_{-\infty}^{+\infty} x(t+\tau)x(\tau)d\tau$$

Let $x(t)$ be a rectangular pulse of amplitude 1 in the range $2 \leq t \leq 4$.

(a) Compute the convolution of $x(t)$ with itself.

(b) Compute the autocorrelation of $x(t)$.

7. (a) If $x_e[n]$ is the conjugate symmetric part of the sequence $x[n]$, what symmetries do the real and imaginary parts of $x_e[n]$ hold?

   (b) Find the conjugate symmetric part of the sequence $x[n] = je^{j4\pi n}$. Comment on the type of sequence.

8. Determine which of the following systems are invertible?

   (a) $y[n] = 2x[n]$

   (b) $y[n] = nx[n]$

   (c) $y[n] = x[n] - x[n-1]$

   (d) $y[n] = Re\{x[n]\}$

9. The input output relation of a system is given by $y[n] = ay[n-1]+x[n]$. Consider the input to the system $x[n] = K\delta[n]$ and the auxillary condition $y[-1] = 3$.

   (a) Find the expression for $y[n]$

   (b) Determine whether the system is LTI?

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